

**H2020-EUJ-02-2016**

H2020 Grant Agreement Number 723076

NICT Management Number 18302

**Deliverable D2.1**

# **Requirements Specification (V1)**

**Version V1.0**

**September 30, 2016**

## **ABSTRACT**

This deliverable specifies an initial set of requirements for the applications that will build on top of the CPaaS.io platform. The objective of this deliverable is (1) to define and communicate what needs to be implemented for each application, and (2) to provide input to the task of defining the platform requirements. Consequently, it contains an initial description of each application that provides a high level understanding of the applications and a high level architecture that shows potential interactions with the CPaaS.io platform. Where available we also provide a description of a user interface. For each application we describe the key data sets and their characteristics as well as a deployment plan that provides an initial overview of how, when and where the application will be deployed. The actual requirements are provided as an initial backlog of user stories as known from agile software development such as Scrum. These requirements are written from an application perspective, i.e. the user stories represent functional and partially non-functional requirements that must be met for implementing the whole application. In a second step, the user stories will – as part of the platform requirements analysis – be fed into the process of identifying the actual subset of requirements for the platform.

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| <b>Delivery Type</b>             | R                                                                        |
| <b>Dissemination Level</b>       | Public                                                                   |
| <b>Contractual Delivery Date</b> | 30/09/2016                                                               |
| <b>Actual Delivery Date</b>      | 14/10/2016                                                               |
| <b>Keywords</b>                  | Requirements, applications, data source                                  |

## Revision History

| Rev. | Date       | Description                                                                                                                              | Contributors             |
|------|------------|------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| 0.1  | 26/07/2016 | TOC and document structure                                                                                                               | Martin Strohbach (AGT)   |
| 0.2  | 04/08/2016 | Update according to meeting: renamed use cases to application, added responsibilities as known, provided template for user story backlog | Martin Strohbach (AGT)   |
| 0.3  | 31/08/2016 | Application description Enhanced User Experience (Section 2)                                                                             | Martin Strohbach (AGT)   |
| 0.4  | 16/09/2016 | Architecture and data sets for Enhanced User Experience (Section 2)                                                                      | Martin Strohbach (AGT)   |
| 0.6  | 19/09/2016 | Section 5 added                                                                                                                          | Alexander Overtoom (TTN) |
| 0.7  | 20/09/2016 | Deployment plan for Enhanced User Experience added (Section 2)                                                                           | Martin Strohbach (AGT)   |
| 0.8  | 22/09/2016 | Initial version of Enhanced User Experience requirements (Appendix A)                                                                    | Martin Strohbach (AGT)   |
| 0.9  | 23/09/2016 | Initial version of Abstract                                                                                                              | Martin Strohbach (AGT)   |
| 0.10 | 28/09/2016 | TTN deployment plan added<br>User requirements added                                                                                     | Alexander Overtoom (TTN) |
| 0.11 | 28/09/2016 | Introduction added                                                                                                                       | Martin Strohbach (AGT)   |
| 0.12 | 29/09/2016 | Summary and introduction for appendix added                                                                                              | Martin Strohbach (AGT)   |
| 0.13 | 30/09/2016 | Amended Waterproof Amsterdam summary in introduction                                                                                     | Martin Strohbach (AGT)   |
| 0.14 | 30/09/2016 | Integrated Tokyo Public transportation and Yokosuka Emergency Medical Care                                                               | Martin Strohbach (AGT)   |
| 0.15 | 1/10/2016  | Appendix for Tokyo Public Transportation and Yokosuka Emergency Medical Care added                                                       | Noboru Koshizuka (UoT)   |
| 0.16 | 1/10/2016  | Tokyo Public Transportation Scenario modified. This has two scenarios, added in independent sections.                                    | Noboru Koshizuka (UoT)   |
| 0.17 | 1/10/2016  | Sapporo Scenario by MSJ integrated                                                                                                       | Noboru Koshizuka (UoT)   |
| 0.18 | 4/10/2016  | Review of version 0.17 with comments and suggestions                                                                                     | Marianne Fraefel (BFH)   |
| 0.19 | 05/10/2016 | Minor comments                                                                                                                           | Martin Strohbach (AGT)   |
| 0.20 | 05/10/2016 | Replied to comments in intro section                                                                                                     | Martin Strohbach (AGT)   |
| 0.21 | 06/10/2016 | Addressed Toshihiko Yamakami review recommendations                                                                                      | Martin Strohbach (AGT)   |
| 0.22 | 06/10/2016 | Added a definition of data set fields in the introduction. Refined value perspective templates in each section.                          | Martin Strohbach (AGT)   |
| 0.23 | 07/10/2016 | Addressed reviewer comments in Section 2. Refactored time plan into deployment plan in each section.                                     | Martin Strohbach (AGT)   |
| 0.24 | 10/10/2016 | Addressed reviewers comments in section 6<br>Added proviso to introduction of section 6                                                  | Alexander Overtoom (TTN) |
| 1.0  | 14/10/2016 | Final edits                                                                                                                              | Martin Strohbach (AGT)   |

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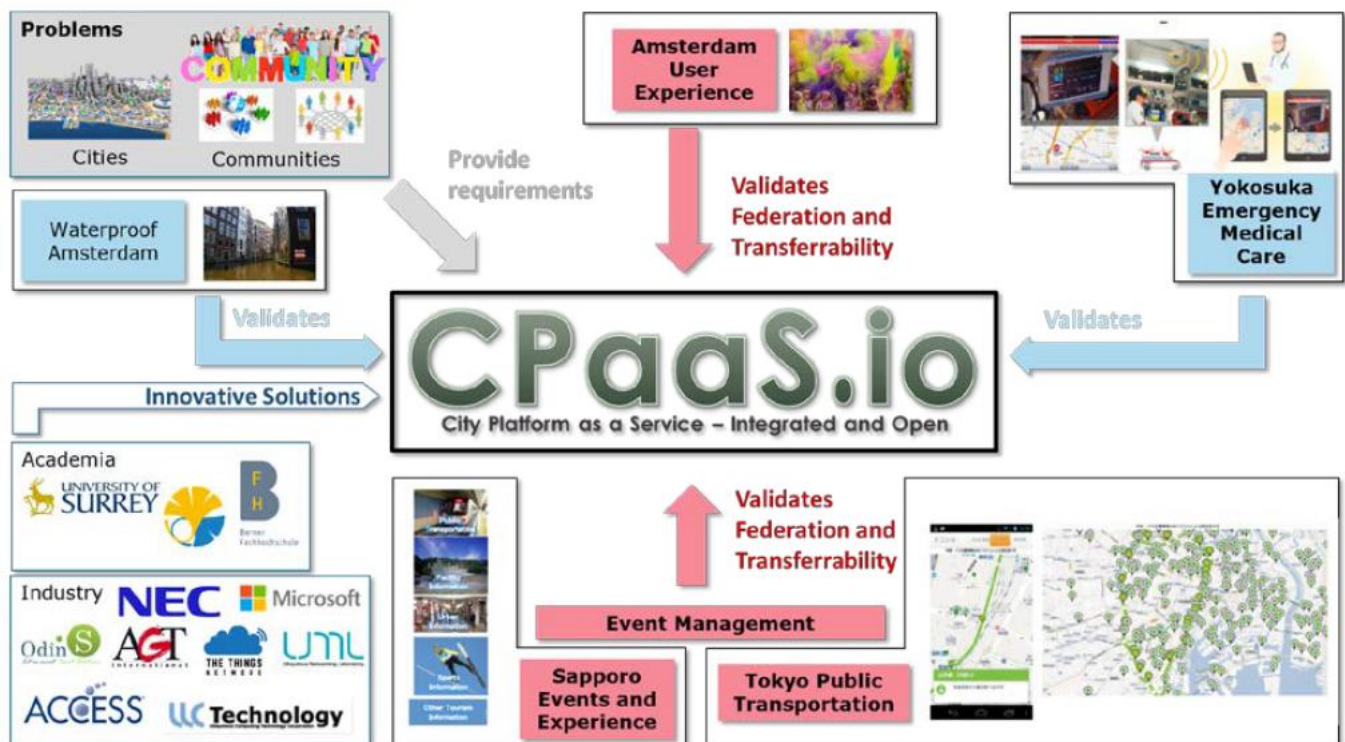
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## Table of Acronyms

| Acronym | Definition             |
|---------|------------------------|
| LOD     | Linked Open Data       |
| ML      | Machine Learning       |
| OGD     | Open Governmental Data |
| WP      | Work package           |

# 1 Introduction

In this deliverable we provide a set of initial use cases and their requirements. Throughout the project we will use these scenarios to demonstrate how applications can benefit from the CPaaS.io platform. These scenarios are carefully chosen to validate that the platform can indeed support a range of diverse usage scenarios that are of societal and commercial value. Consequently, the scenarios are used as input for the requirement analysis of the platform as carried out in WP3. In the validation phases we will use the application to demonstrate the federation and transfer capabilities of the platform.



**Figure 1:** Overview of the CPaaS.io validation concept. Cities and local as well as virtual communities provide problems defining the requirements. The consortium provides innovative technical solutions that are implemented as part of the platform. The use case validates the platform including the transferability between Japan and Europe. (This picture is taken from the Grant Agreement document).

As depicted in Figure 1 we will implement 3 use cases and 5 application scenarios:

- Use Case 1: Managing Fun and Sports Events.** This use case addresses event management challenges associated both with the supporting city infrastructure such as transportation as well as with managing the event on site and enhancing the visitor and user experience. It is used as the driving use case for the platform validation and federation capabilities and will need to fully draw on the technical features of the platform such as citizen empowerment, cloud and edge computing, and holistic data management. In order to demonstrate not only the cross-regional, but also cross-application use of data, three of the application scenarios will be implemented to realize this complex use case. The *Enhanced User Experience* application in Section 2 focuses on sensor intensive analytics to enhance people's experience and will first be deployed in Europe. The *Visitor Experience* application in Section 3 will both address logistic challenges related to large events and provide visitor services for these events. Finally, the *Public Transportation* application will provide next generation mobile transportation for all citizens. This application is further split into two sub-applications. Section 4 details an application that integrates available open public transportation data and Section 5 includes vehicle such as taxis, trucks, and utility vehicles by public and private

companies. Both the *Visitor Experience* and *Public Transportation* application will initially be deployed in Japan.

- **Use Case 2: Waterproof Amsterdam.** This use case emphasizes the diversity of city applications supported by the platform. It contains a single, but highly relevant application that is concerned with the problem that increasing amount of rainfall in Amsterdam is overloading the capacities of the sewage system. This use case further poses some unique demands to the platform in that it operates an actuator mechanism.
- **Use Case 3 Yokosuka Emergency Medical.** This is the third use case that further will demonstrate how IoT technologies used across the city help to maintain health and welfare of citizens. Specifically, the applications developed in this use case will aid first aiders to obtain life-saving and relevant information when and where they are needed.

This deliverable introduces all five application scenarios in their own sections. For each application scenario we provide an initial description of each application, a high level architecture, where available a user interface, a list of data sets and a deployment plan. The architecture sketches an initial idea of platform interactions and is meant to be quickly iterated in the platform requirements analysis phase.

The data set descriptions further characterize the application and provide important information with respect to the required data management functionalities of the platform. The deployment plan provides an initial overview of how, when and where the application will be deployed and helps to understand the schedule of the individual application deployments and their complexity.

**Table 1:** Description of the fields used in the data set descriptions

| Data Set Name               | A name that identifies the data set                                                                                                                                                                                                                                                                                  |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Detailed Description</b> | A verbal description of the data set.                                                                                                                                                                                                                                                                                |
| <b>OGD or private data</b>  | Classifies the data whether it is open governmental data (OGD) or private data, i.e. data that is not provided by governmental institutions. Possible values are <i>OGD</i> or <i>Private data</i> .                                                                                                                 |
| <b>Personal Data</b>        | Indicates whether information contained in the data set identifies or could be used to identify any individual person. Possible values are <i>yes</i> and <i>no</i> .                                                                                                                                                |
| <b>Hosting</b>              | Indicates whether the data set is planned to be hosted by the CPaaS.io platform or not. Possible values are <i>platform</i> and <i>external</i> .                                                                                                                                                                    |
| <b>Data Provider</b>        | Describes the entity that is offering the data set. This can be an open data portal, a partner of the consortium or any other private and public entity that provides the data. Possible values include a string that identifies the data provider such as partner name, city name/department and may include a URL. |
| <b>Format</b>               | Describes the technical format in which the data is provided. Possible values include a string referencing a technical format. Examples include CSV for comma separate value or RDF for a serialization format for data represented in the Resource Description Format.                                              |
| <b>Update Frequency</b>     | Describes the rate in which updates of the data sets are available. This may be specified in Hz or by specifying a time interval between an average update, e.g. 1 update per hour, minute or year etc.                                                                                                              |
| <b>Update Size</b>          | Describe how big a single update. Possible values include bytes, Kilobytes, etc. or a number of atomic data units such as a record, document, or triple.                                                                                                                                                             |
| <b>Data Source</b>          | Technically characterizes the source. Possible values include but are not limited to <i>SQL database</i> , <i>sensor</i> , <i>file</i> .                                                                                                                                                                             |

| Data Set Name            | A name that identifies the data set                                                                                                                                                                                                  |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sensor</b>            | Describes the sensor that provides the data if applicable. Possible values are <i>N/A</i> if the data source is not a sensor or a string describing the sensor including but not limited to <i>wristband</i> , <i>mobile phone</i> , |
| <b>Number of Sensors</b> | The amount of sensors used in a deployment if applicable. The field may be extended to specify a finer deployment granularity, e.g <i>number of sensors per person</i> . Possible values are integer values.                         |

The appendix contains the list of requirements in the form of an initial backlog of user stories as known from agile software development such as Scrum. These requirements are written from an application perspective, i.e. the user stories represent functional and partially non-functional requirements that are necessary for implementing the whole application. In a second step, the user stories will – as part of the platform requirements analysis – be fed into the process of identifying the actual subset of requirements on the platform.

## 2 Event Management - Enhanced User Experience

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The *Enhanced User Experience application* is one of the applications that are part of the *Managing Fun and Sports Events Use Case* or short *Event Management*. The *Enhanced User Experience* application aims at providing the best possible experiences to event participants. It is a sensor intensive application that draws from wearables and other sensors with the objective to make event visits for participants unique.

### 2.1 Application Description

The core idea of this application is to use IoT sensors and analytics to enhance people's experience while visiting or participating at a fun or sports event. Wearables and mobile phones are used as sensors in order to learn about the activities of event participants. Event participants may include members of the audience, but also performing artists or athletes. For instance AGT has previously equipped referees and cheerleaders in basketball matches with wearable sensors and created content based on the analysed data for consumption on site and for distribution via TV broadcasting, social media and other digital distribution channels<sup>1</sup>. Furthermore the application uses sensor deployed at the venue to measure and analyse fan behaviour and engagement.

We will extend this work and initially focus on active participants that are equipped with wearables. As a driving event, we plan to deploy our sensors and analytics at Color Run events<sup>2</sup>. The Color Run is a global event series that takes place in many cities around the world including Tokyo and Amsterdam. At the Color Run, participants run a distance of five kilometres and are being sprayed with color on their way. As the event focuses on health and fun rather than athletic achievements the aim for the participants is to reach the finish line rather than comparing the running time with other runners. At the finish line the runners gather to party and dance to music getting sprayed with more color.

AGT will draw from its analytics portfolio to visualize the user's personal data for each event. For instance, wristband data can be used to show trajectories, effort (heartbeat) and excitement levels during the events

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<sup>1</sup><http://www.euroleague.net/final-four/berlin-2016/news/i/6vokoibj5fsgqg4q/heed-the-event-platform-based-joint-venture-between-wme-img-and-agt-international-makes-first-official-foray-into-sports>

<sup>2</sup> <http://thecolorrun.com/>

and video analytics solutions can be used to create dynamic, geo-linked colour maps that can be linked to runners' "sensor trails".

The frequency of events in both Europe and Japan renders this application useful to validate the federation concepts of the CPaaS.io platform. As this application relies on the collection and management of personal data it is paramount that it protects the privacy of the event participants and uses CPaaS.io's citizen empowerment features developed in WP 5.

### 2.1.1 Value Perspective

In the following tables we describe two values perspective. Table 2 lists the various user and stakeholder groups that benefit from the application and list for each group the benefits they have by using the application. Table 3 in turns takes the perspective from the platform and lists for each key concept the potential benefit the CPaaS.io platform can provide to the application.

**Table 2:** The benefits the application provides to the different user groups

| User Value Perspective    |                                                   |
|---------------------------|---------------------------------------------------|
| <b>Event Participants</b> | Recorded personal experiences of visited events   |
| <b>Event Organizers</b>   | Value-added services attracting more participants |

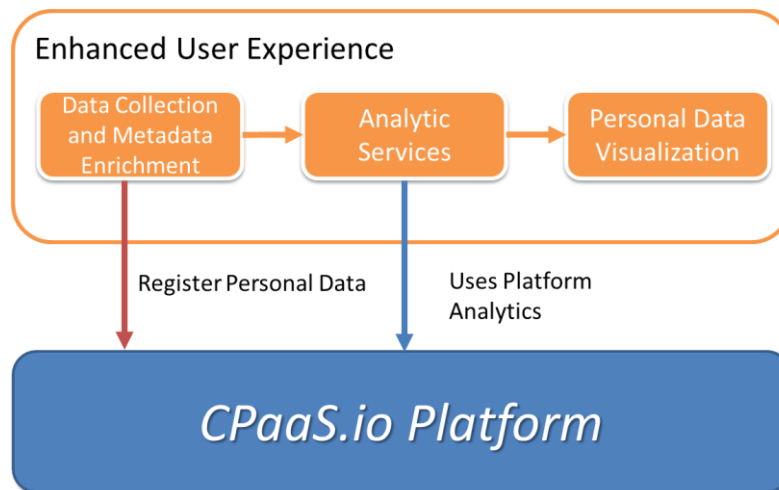
**Table 3:** The benefits the CPaaS.io platform provides to the application

| Platform Value Perspective           |                                                                                               |
|--------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Use of IoT Technologies</b>       | Data collection and integration of wearable sensor data, real-time analysis of sensor streams |
| <b>Use of Open Governmental Data</b> | Integration of geospatial data for user visualizations                                        |
| <b>Use of Linked Data</b>            | Metadata descriptions of sensors, integration with related data sets                          |
| <b>Big Data Analysis</b>             | Re-use of ML and analysis toolkits                                                            |
| <b>My Data Approach</b>              | Transparency and control over use of personal data                                            |

## 2.2 Architecture

Figure 2 shows an initial version of the high level components and their envisioned interaction with the CPaaS.io platform. It contains the following components

- **Data Collection** is responsible for collecting all data necessary for this application as detailed in Section 2.3. This includes primarily collecting data from wearables and other sensor deployed at the event, but may also include other data from third parties such as meta information about the event. Other metadata includes data related to the deployment that will be retrieved from the deployment knowledge base developed in WP6 and required user information. The data collection component registers data at the CPaaS.io platform, but may also retrieve data from the platform provided by other applications or data providers, e.g. data providers for the event metadata.
- **Analytic Services** will use raw data from wearables, environmental sensors and other data to compute personal event data. The component encapsulates various data aggregation and machine learning modules.
- **Personal Data Visualization** is the front end component that offers users to view their data per event using different visualizations. This front end should provide means for controlling what data is collected and how (e.g. update frequency). In particular it should provide feedback about the minimum settings for data collection that is required for presenting useful analytics results.



**Figure 2:** Enhanced User Experience: Initial version of the high level application architecture.

## 2.3 User Interface

The user interface will be provided as a web interface that provides access to personalized visualizations of individuals. There will also be a mobile version, either as mobile web application or as native mobile app.

## 2.4 Data Sets

In this section we briefly describe the data sets that will be used in this application. This list reflects the initial data sets that we plan to use for this application and is therefore subject for change.

The data described here is used for analytics and generating personalized visualizations. While all of the data described below constitutes personal data collected primarily from wearables not every data set is used for each users to the same extent. Whether a data set is stored and used for processing, depends on several factors, such as

- **Sensor Deployment:** Has the respective sensor been deployed? E.g. did the event participant wear a device that has a sensor for measuring heart beat information?
- **User Control:** Did the event participant allow the recording of the respective data in the way described in the data set? The application foresees that the end user has full control over what data is collected and in what way. This means that a user could decide to not collect heart beat data, or lower the update frequency to a few seconds. Of course this may have implications on the information that can be derived from the data.
- **Data Quality:** Is the data of sufficient quality in order to be presented to the user? Among other factors data quality can be assessed by having knowledge about the sensor accuracy, the update frequency or data set size. For instance wearables differ widely in the accuracy of their heart rate sensors. The data set size can have an impact on providing optimised models for machine learning or may not cover relevant time spans.

### Biometric data

#### Detailed Description

We will collect a range of biometric measurements from wearables such as wristbands, chest straps and smart sportswear that provides biometric measurements including heart rate, breathing rate and galvanic skin response, burned calories measurements and skin temperature.

| Biometric data                      |                                      |
|-------------------------------------|--------------------------------------|
| <b>OGD or private data</b>          | Private                              |
| <b>Personal Data</b>                | Yes                                  |
| <b>Hosting</b>                      | External                             |
| <b>Data Provider</b>                | AGT                                  |
| <b>Format</b>                       | JSON                                 |
| <b>Update Frequency</b>             | up to every 200ms                    |
| <b>Update Size</b>                  | ~1 KB                                |
| <b>Data Source</b>                  | Sensor                               |
| <b>Sensor</b>                       | Wristband, chest strap, smart shirts |
| <b>Number of Sensors per person</b> | ~6                                   |

| GPS Traces                          |                                                                                                |
|-------------------------------------|------------------------------------------------------------------------------------------------|
| <b>Detailed Description</b>         | GPS traces include positional data including altitude information as delivered by GPS devices. |
| <b>OGD or private data</b>          | Private                                                                                        |
| <b>Personal Data</b>                | Yes                                                                                            |
| <b>Hosting</b>                      | External                                                                                       |
| <b>Data Provider</b>                | AGT                                                                                            |
| <b>Format</b>                       | common GPS formats (GPX, KML, CSV, NMEA)                                                       |
| <b>Update Frequency</b>             | Up to 1s                                                                                       |
| <b>Update Size</b>                  | < 1KB                                                                                          |
| <b>Data Source</b>                  | Sensor                                                                                         |
| <b>Sensor</b>                       | GPS sensor in wristbands and mobile phones                                                     |
| <b>Number of Sensors per person</b> | 1-2                                                                                            |

| Motion Data                         |                                                                                                |
|-------------------------------------|------------------------------------------------------------------------------------------------|
| <b>Detailed Description</b>         | Motion data that measures hand and body movements based on accelerometer and gyroscope sensors |
| <b>OGD or private data</b>          | Private                                                                                        |
| <b>Personal Data</b>                | Yes                                                                                            |
| <b>Hosting</b>                      | External                                                                                       |
| <b>Data Provider</b>                | AGT                                                                                            |
| <b>Format</b>                       | JSON                                                                                           |
| <b>Update Frequency</b>             | Up to every 16 ms                                                                              |
| <b>Update Size</b>                  | ~ 200 byte per sensor reading                                                                  |
| <b>Data Source</b>                  | Sensors                                                                                        |
| <b>Sensor</b>                       | Accelerometer and gyroscope sensors of mobile phones, wristband and other wearables            |
| <b>Number of Sensors per person</b> | 2-3                                                                                            |

| Step Counts                 |                                     |
|-----------------------------|-------------------------------------|
| <b>Detailed Description</b> | This data set contains step counts. |

| Step Counts                         |                                     |
|-------------------------------------|-------------------------------------|
| <b>OGD or private data</b>          | Private                             |
| <b>Personal Data</b>                | Yes                                 |
| <b>Hosting</b>                      | External                            |
| <b>Data Provider</b>                | AGT                                 |
| <b>Format</b>                       | JSON                                |
| <b>Update Frequency</b>             | Up to 1Hz                           |
| <b>Update Size</b>                  | ~ 200 byte per sensor reading       |
| <b>Data Source</b>                  | Sensors                             |
| <b>Sensor</b>                       | Step count measurement of wristband |
| <b>Number of Sensors per person</b> | 1-2                                 |

| Environmental Data                  |                                                                                                                                       |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Detailed Description</b>         | This data set environmental data such light intensity and barometric pressure. The data is primarily collected from wearable sensors. |
| <b>OGD or private data</b>          | Private                                                                                                                               |
| <b>Personal Data</b>                | Yes (tbc)                                                                                                                             |
| <b>Hosting</b>                      | External                                                                                                                              |
| <b>Data Provider</b>                | AGT                                                                                                                                   |
| <b>Format</b>                       | JSON                                                                                                                                  |
| <b>Update Frequency</b>             | Up to 1Hz                                                                                                                             |
| <b>Update Size</b>                  | ~ 200 byte per sensor reading                                                                                                         |
| <b>Data Source</b>                  | Sensors                                                                                                                               |
| <b>Sensor</b>                       | Sensors in wristband                                                                                                                  |
| <b>Number of Sensors per person</b> | 1-2                                                                                                                                   |

## 2.5 Deployment Plan

In the simplest case a deployment of the enhanced user experience application consists of equipping participants with wearables that provide their data in real-time to the data collection platform for further processing. Additionally an event site or venue may be equipped with further sensors such as cameras or microphones.

AGT has already deployed sensors on a group of 8 participants during the Color Run in Utrecht, The Netherlands on 11<sup>th</sup> September 2016. The purpose of this deployment was to compare wearable device technologies suitable for the event type and validating the analytics. Data using the sensors described above has been used for both during the 5 kilometre run and the after run party. For obtaining ground truth for our biometric data AGT also used video recordings of the 8 participants. The data collected at this event will provide the basis to develop the initial set of services for the first prototype (Deliverable D2.2, due December 2016).

The enhanced user experience application will be iteratively developed and incrementally use platform components and concepts. In alignment with the project plan we are planning to deploy and evaluate the first prototype at another Color Run event early next year. For instance, we are considering the Color Run

in Amsterdam that will take place on the 28<sup>th</sup> of May 2017. The plan for developing the Enhanced User Experience application in the first year is as follows:

- M01-M03: Initial requirement analysis and 1<sup>st</sup> deployment in order to collect data (D2.1)
- M03-M06: Data analysis and implementation of first analytics prototype (D2.2)
- M06-M12: 2<sup>nd</sup> deployment including test of analytics chain (internal milestone)

For the second and following prototypes we would like to gradually extend the application to other event types and locations in Japan.

### 3 Event Management – Sapporo Visitor Experience

---

The *Sapporo Visitor Experience application* is part of a set of applications that will use the CPaaS.io platform to manage fun and sports event. This application focuses on tourist information services including event information and public transportation information.

#### 3.1 Application Description

We deliver mobile applications “Kokosil Sapporo” that provide information of sightseeing spots, restaurants, hotels, public transportation and events for tourists. By leveraging a geo-spatial position recognition system, tourists can get information at the point where they need it in a timely manner. They can drill down on spot information by tapping an icon on the map made available in the mobile application. By tapping the “go to this place” button, a navigation service from where they are to the place of interest is made available on Google Maps. Since information is translated in multiple languages, users can select a language in the language settings for their mobile devices.

Sapporo Snow Festival Guide is one of the mobile applications that provide event information to visitors. The Sapporo Snow Festival is a very large event that is visited by 3 million people in February every year. This application will deliver relevant event information such as location and additional information on snow sculptures, shops, toilets, and tour bus meeting points.

We are going to develop an application for The Asian Winter Games held in February 2017 as well. This will provide information on the game schedule and locations, rules, stadium facilities and transportation services.

#### 3.2 Architecture

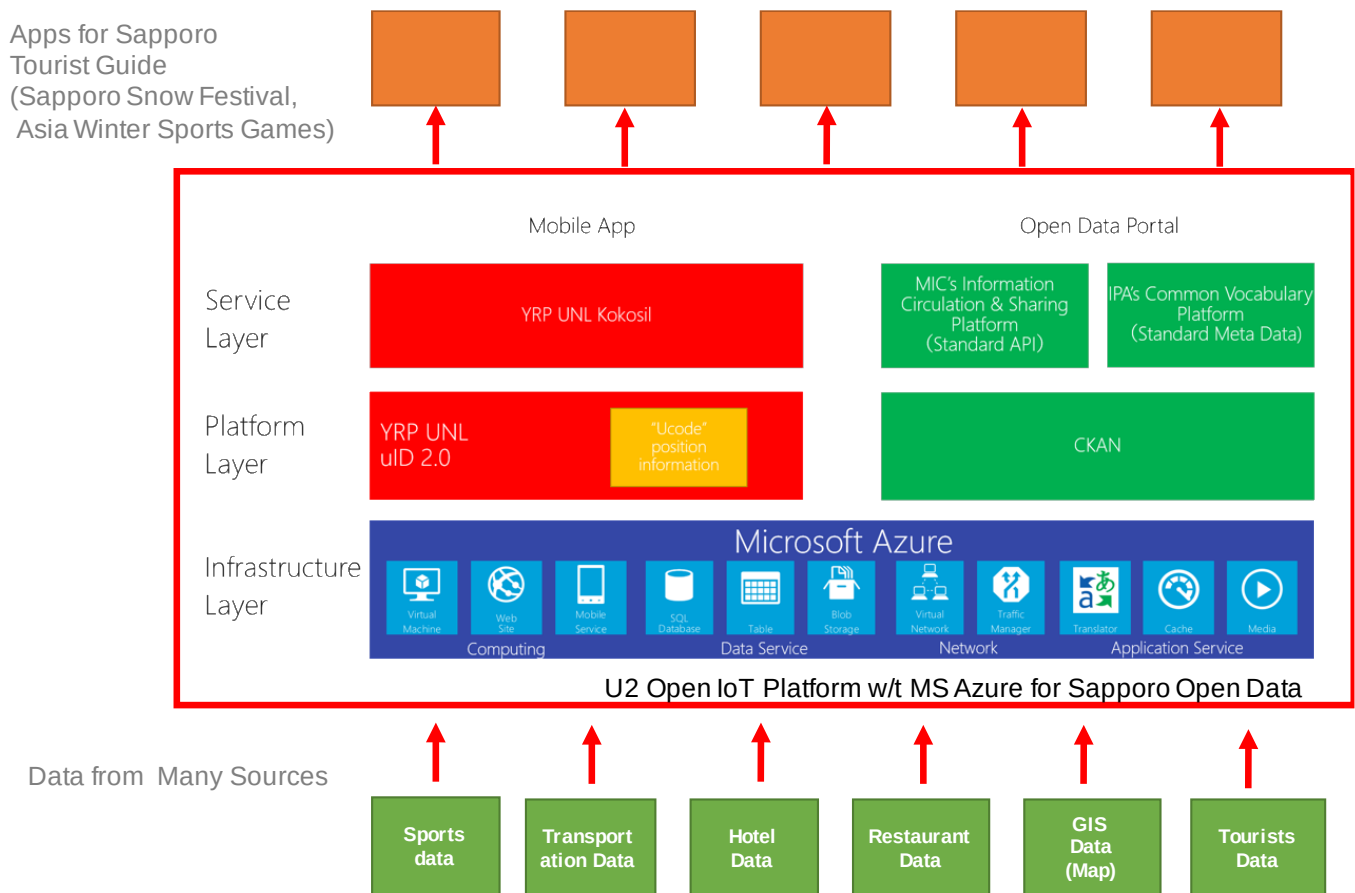
Figure 3 shows an initial version of the high level components and their envisioned interaction with the CPaaS.io platform. It contains the following components:

- The **Open Data Portal** is built on CKAN and Microsoft Azure. The platform provides Open Data as well as guidance for application developers. Developers can use industry standard tools including an API defined by the Ministry of Internal Affairs and Communications and a vocabulary defined by the Information-technology Promotion Agency.
- **Mobile applications** leveraging open data are implemented on YRP UNL’s Kokosil and uID 2.0. Kokosil receives a ucode<sup>3</sup> as the position ID from the BLE (Bluetooth Low Energy) markers installed at information signage. Then Kokosil sends the ucode to uID 2.0, which manages the ucode system

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<sup>3</sup> “ucode” is an ID system approved as ITU-T’s International Standards H.642.1.

and related data, and gets information connected with the ucode. This mechanism can recognize the position of the Kokosil user more precisely than GPS in particular, whether it is indoor or outdoor. The information provided by Kokosil is automatically translated to multiple languages by Microsoft Translator.



**Figure 3:** Visitor Experience: Initial version of the high level application architecture.

### 3.3 User Interface

The user interface will be provided as mobile web applications (Kokosil) for iOS and Android devices. Those screen shots below are sample user interface.

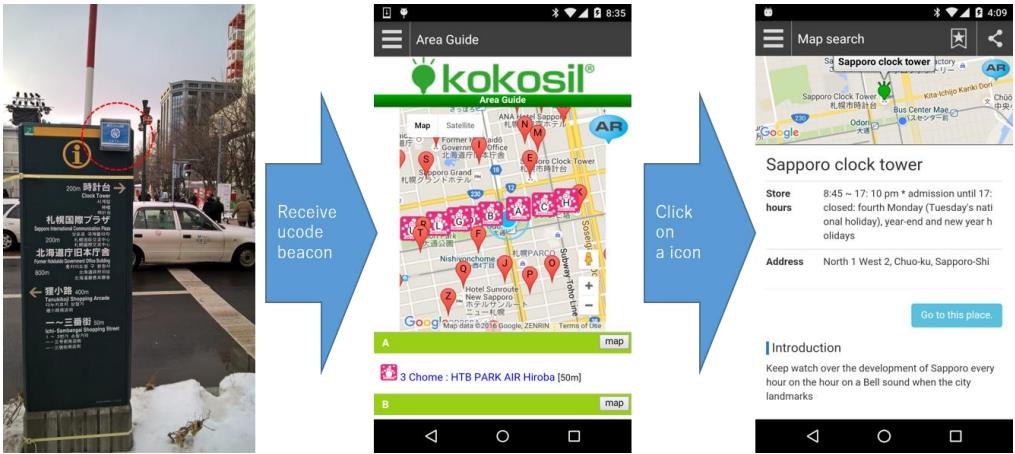


Figure 4: Tourist information service with geo spatial position recognition



Figure 5: Sports event information



**Figure 6:** Public Transportation Information

### 3.4 Data Sets

We established the “Sapporo Open Data Association”, whose members are data providers such as Sapporo City, companies and organizations in tourism, sports and transportations. The association collects data from its members, transforms them to Open Data and publishes them on the Sapporo Open Data Portal.

| Sightseeing Spot            |                                                               |
|-----------------------------|---------------------------------------------------------------|
| <b>Detailed Description</b> | Comments, pictures and locations of popular sightseeing spots |
| <b>OGD or private data</b>  | OGD                                                           |
| <b>Personal Data</b>        | No                                                            |
| <b>Hosting</b>              | Sapporo Open Data Association                                 |
| <b>Data Provider</b>        | Sapporo City                                                  |
| <b>Format</b>               | PDF, HTML, CSV, JPEG                                          |
| <b>Update Frequency</b>     | Yearly                                                        |
| <b>Update Size</b>          | N/A                                                           |
| <b>Data Source</b>          | Web                                                           |
| <b>Sensor</b>               | Ucode marker                                                  |
| <b>Number of Sensors</b>    | 11                                                            |

| Meal                        |                                                         |
|-----------------------------|---------------------------------------------------------|
| <b>Detailed Description</b> | Restaurant name, location, opening hours, menu, service |
| <b>OGD or private data</b>  | Private                                                 |
| <b>Personal Data</b>        | No                                                      |
| <b>Hosting</b>              | Sapporo Open Data Association                           |
| <b>Data Provider</b>        | Hotel, Sapporo Station Shopping Mall                    |
| <b>Format</b>               | PDF, HTML, CSV, JPEG                                    |
| <b>Update Frequency</b>     | Monthly                                                 |
| <b>Update Size</b>          | N/A                                                     |
| <b>Data Source</b>          | Web                                                     |
| <b>Sensor</b>               | Ucode marker                                            |
| <b>Number of Sensors</b>    | 11                                                      |

### Hotel

|                             |                                               |
|-----------------------------|-----------------------------------------------|
| <b>Detailed Description</b> | Hotel name, location, number of rooms, access |
| <b>OGD or private data</b>  | Private                                       |
| <b>Personal Data</b>        | No                                            |
| <b>Hosting</b>              | Sapporo Open Data Association                 |
| <b>Data Provider</b>        | Hotels                                        |
| <b>Format</b>               | PDF, HTML, CSV, JPEG                          |
| <b>Update Frequency</b>     | Monthly                                       |
| <b>Update Size</b>          | N/A                                           |
| <b>Data Source</b>          | Paper                                         |
| <b>Sensor</b>               | N/A                                           |
| <b>Number of Sensors</b>    | N/A                                           |

| Other Facilities            |                                              |
|-----------------------------|----------------------------------------------|
| <b>Detailed Description</b> | Location of ATM, Toilet, Rental Locker, etc. |
| <b>OGD or private data</b>  | Private                                      |
| <b>Personal Data</b>        | No                                           |
| <b>Hosting</b>              | Sapporo Open Data Association                |
| <b>Data Provider</b>        | Sapporo Station Shopping Mall                |
| <b>Format</b>               | PDF, HTML, CSV, JPEG                         |
| <b>Update Frequency</b>     | Monthly                                      |
| <b>Update Size</b>          | N/A                                          |
| <b>Data Source</b>          | Web                                          |
| <b>Sensor</b>               | Ucode marker                                 |
| <b>Number of Sensors</b>    | 11                                           |

| Snow Festival               |                                              |
|-----------------------------|----------------------------------------------|
| <b>Detailed Description</b> | Event information in Sapporo Snow Festival   |
| <b>OGD or private data</b>  | Private                                      |
| <b>Personal Data</b>        | No                                           |
| <b>Hosting</b>              | Sapporo Open Data Association                |
| <b>Data Provider</b>        | Operating Committee of Sapporo Snow Festival |
| <b>Format</b>               | PDF, HTML, CSV, JPEG                         |
| <b>Update Frequency</b>     | Yearly                                       |
| <b>Update Size</b>          |                                              |
| <b>Data Source</b>          | Web                                          |
| <b>Sensor</b>               | Ucode marker                                 |
| <b>Number of Sensors</b>    | 11                                           |

| Stadium Data                |                                              |
|-----------------------------|----------------------------------------------|
| <b>Detailed Description</b> | Location, facilities and drawing of stadiums |
| <b>OGD or private data</b>  | OGD                                          |
| <b>Personal Data</b>        | No                                           |
| <b>Hosting</b>              | Sapporo Open Data Association                |
| <b>Data Provider</b>        | Sapporo City                                 |
| <b>Format</b>               | PDF, HTML, CSV, JPEG                         |
| <b>Update Frequency</b>     | Yearly                                       |

|                          |            |
|--------------------------|------------|
| <b>Update Size</b>       | N/A        |
| <b>Data Source</b>       | Web, Paper |
| <b>Sensor</b>            | N/A        |
| <b>Number of Sensors</b> | N/A        |

| <b>Train &amp; Bus Data</b> |                                                                        |
|-----------------------------|------------------------------------------------------------------------|
| <b>Detailed Description</b> | Time table, Operating status, Line, Real time location, bus stop, etc. |
| <b>OGD or private data</b>  | OGD & private                                                          |
| <b>Personal Data</b>        | No                                                                     |
| <b>Hosting</b>              | Sapporo Open Data Association                                          |
| <b>Data Provider</b>        | Sapporo City, Hokkaido Chuo Bus                                        |
| <b>Format</b>               | PDF, HTML, CSV, JPEG                                                   |
| <b>Update Frequency</b>     | Monthly                                                                |
| <b>Update Size</b>          | N/A                                                                    |
| <b>Data Source</b>          | File                                                                   |
| <b>Sensor</b>               | N/A                                                                    |
| <b>Number of Sensors</b>    | N/A                                                                    |

| <b>Express Way Data</b>     |                               |
|-----------------------------|-------------------------------|
| <b>Detailed Description</b> | Information of service area   |
| <b>OGD or private data</b>  | Private                       |
| <b>Personal Data</b>        | No                            |
| <b>Hosting</b>              | Sapporo Open Data Association |
| <b>Data Provider</b>        | NEXCO East Japan              |
| <b>Format</b>               | PDF, HTML, CSV, JPEG          |
| <b>Update Frequency</b>     | Yearly                        |
| <b>Update Size</b>          | N/A                           |
| <b>Data Source</b>          | Web                           |
| <b>Sensor</b>               | N/A                           |
| <b>Number of Sensors</b>    | N/A                           |

| <b>Underground Map Data</b> |                               |
|-----------------------------|-------------------------------|
| <b>Detailed Description</b> | Drawing of underground map    |
| <b>OGD or private data</b>  | Private                       |
| <b>Personal Data</b>        | No                            |
| <b>Hosting</b>              | Sapporo Open Data Association |
| <b>Data Provider</b>        | Sapporo Station Shopping Mall |
| <b>Format</b>               | PDF, HTML, CSV, JPEG          |
| <b>Update Frequency</b>     | Yearly                        |
| <b>Update Size</b>          | N/A                           |
| <b>Data Source</b>          | File                          |
| <b>Sensor</b>               | Ucode marker                  |
| <b>Number of Sensors</b>    | 11                            |

### 3.5 Deployment Plan

We will start information services on Kokosil before Sapporo Snow Festival and Asian Winter Games in February 2017. At the event venue, we will introduce the Kokosil service to visitors and collect feedback from users.

## 4 Event Management – Tokyo Public Transportation (1)

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The *Tokyo Public Transport application* is part of a set of applications that will use the CPaaS.io platform to manage fun and sports event. This sub-application focuses on integrating data from different public transportation services.

### 4.1 Application Description

Public transportation data is highly public in nature, and much of the public transportation services are operated by the public sector and private companies. Today, they are providing information services such as transfer guidance services based on timetables, and operational information services such as delays and service troubles. For buses, some operators are providing location information and operational information in real time by so-called bus location systems.

Tokyo and other mega-cities have huge and complicated networks of public transportation consisting of hundreds of railway stations, thousands of bus stops and routes, many operators, and so on. In this case, open data is the best and only approach for providing integrated information about public transportation networks.

On the other hand, for monitoring real-time public transportation information such as location of trains and/or busses, we also need IoT technologies on top of open data. So, integrated public transportation information services are good use cases for using both open data technologies and IoT technologies. For example, in this project, we use an RDF model based on the LOD technology for the data model, the ucode system developed by Ubiquitous ID Center<sup>4</sup>, SPARQL and RESTful API with JSON format, and a special new vocabulary for the expression of public transportation data.

This use case trial shall be performed in Tokyo Area with the support of the "Association of Open Data for Public Transportation" (ODPT)<sup>5</sup>. <http://odpt.org/>. As of January 2016, the association has 39 members.

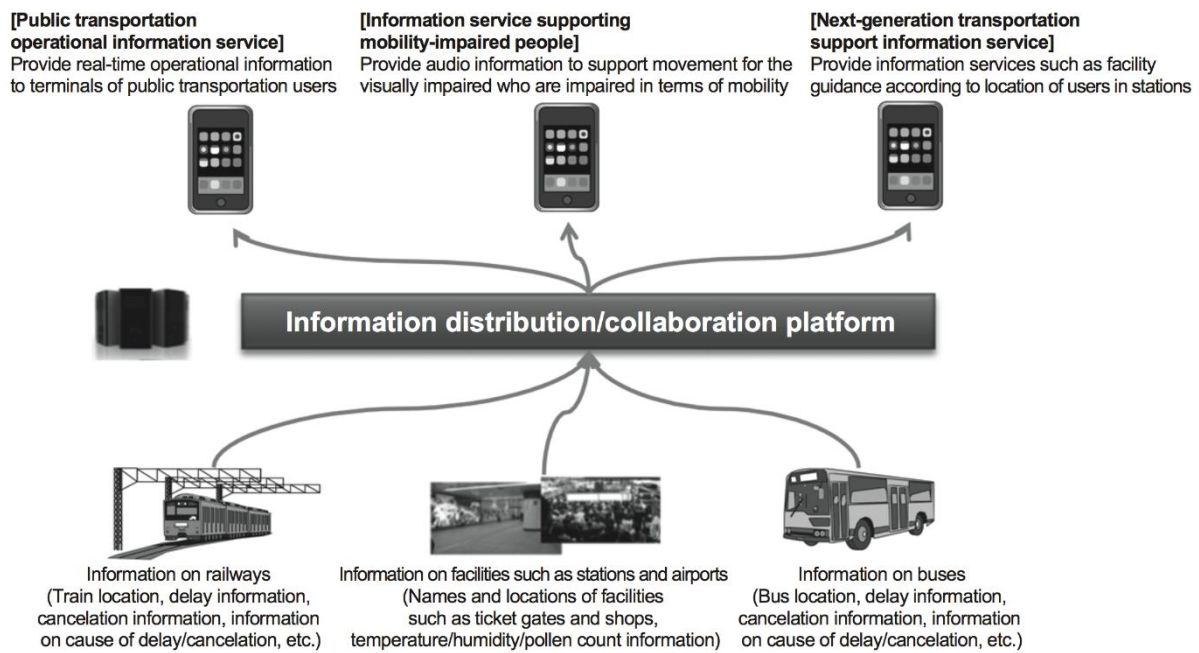
### 4.2 Architecture

The following figure is the architecture plan overview of the public transportation application of CPaaS. We will use the CPaaS.io platform for the "information distribution/collaboration platform" portion in the following figure.

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<sup>4</sup> <http://uidcenter.org>

<sup>5</sup> <http://odpt.org>



**Figure 7:** Public Transportation: Initial version of the high level application architecture.

### 4.3 User Interface

Two user interfaces will be provided: one for end users of the information service and one for developers.

The end-user interface will be provided as a mobile web application. Figure 8 and Figure 9 are prototype images of the end-user interface of our public transportation information services using open data and/or private data.



**Figure 8:** Prototypes of end-user interface of public transportation information services

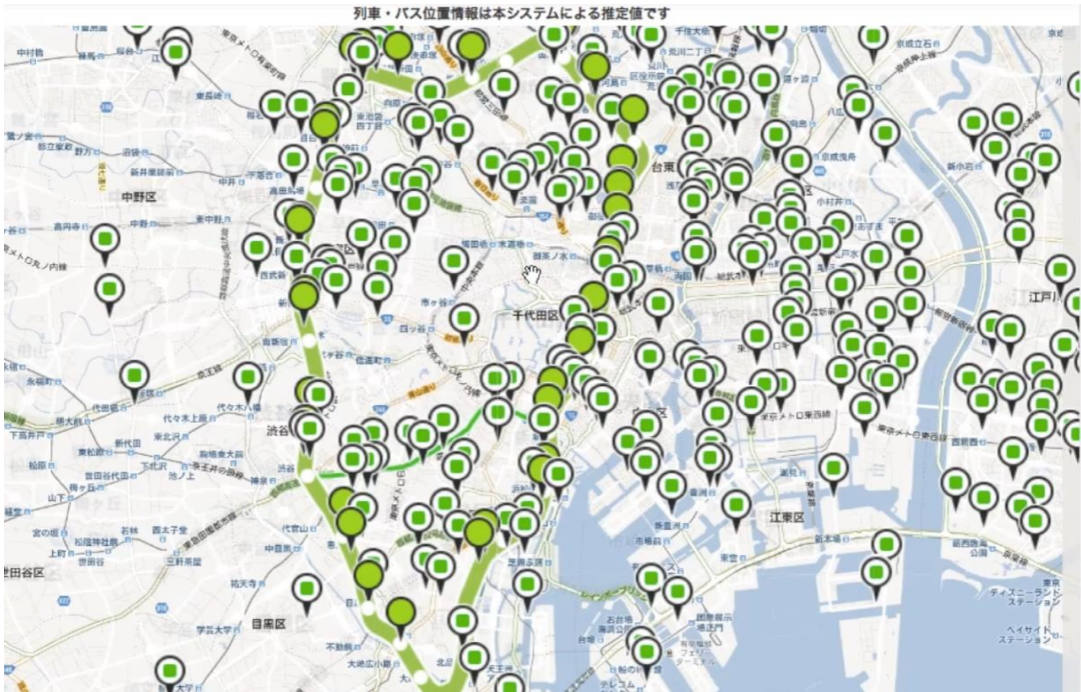


Figure 9: Prototypes of large screen view of end-user interface of public transportation information services

The user interface for software/service developers, will be provided as web interface, i.e. “Developers' site”. Figure 10 shows a prototype of the developers' site.



Figure 10: Prototype of the developer user interface of the public transportation application

4.4 Data Sets

We will collaborate with the “Association for Open Data of Public Transportation”. We will also have support from the Ministry of Land, Infrastructure, Transport and Tourism which releases many kinds of open data for public transportation. In our application, we will use data sets as summarized in the following table.

| Tokyo Public Transportation Data |                                                                                                                                                                                                                  |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Detailed Description             | (1) Time tables and real-time traffic information of public transportation such as railways, buses, and airplanes.<br>(2) Geographical data for routes, locations of railway stations, bus stops, airports, etc. |

| Tokyo Public Transportation Data |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | (3) Data of railway stations such as maps, facility lists,                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>OGD or private data</b>       | OGD and Private Data                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Personal Data</b>             | Not included                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Hosting</b>                   | Consortium of Open Data for Public Transportation                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Data Provider</b>             | Consortium of Open Data for Public Transportation, and many public transportation companies and sectors                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Format</b>                    | ucR (ucode Relation) format (RDF + ucode in JSON format)                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Update Frequency</b>          | Yearly: Geographical data, railway station data, etc.<br>About every 6 months: Static data for operation such as time tables, etc.<br>Real-time update: real-time traffic information, etc.                                                                                                                                                                                                                                                                                     |
| <b>Update Size</b>               | New real-time data is appended to (old) existing data.<br>Yearly update size and 6-month update size are estimated to be relatively small (less than 10% of the existing data size).                                                                                                                                                                                                                                                                                            |
| <b>Data Source</b>               | <ol style="list-style-type: none"> <li>1. Tokyo Public Transportation Open Data<br/><a href="http://www.odpt.org/">http://www.odpt.org/</a></li> <li>2. Tokyo Metro Open Data<br/><a href="https://developer.tokyometroapp.jp/info">https://developer.tokyometroapp.jp/info</a></li> <li>3. Open Data for Pedestrian Navigation Support Services<br/><a href="https://www.hokoukukan.go.jp/top.html">https://www.hokoukukan.go.jp/top.html</a></li> <li>4. 4. others</li> </ol> |
| <b>Sensor</b>                    | Location sensors of trains, GPS for bus locations, etc.                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Number of Sensors</b>         | Hard to count/ hard to estimate (To output the data, all train and bus cars are using sensors.)                                                                                                                                                                                                                                                                                                                                                                                 |

## 4.5 Deployment Plan

We have already been organizing "Association of Open Data for Public Transportation (ODPT)", which has more than 50 members including railway companies, bus companies, Japan Air Line, All Nippon Air Line, ICT companies such as Fujitsu, NEC, Google, and Microsoft. We will collaborate with this association for the deployment of the result of this project.

As the first step, we are planning to start public transportation open data services in Tokyo. The final target year is 2020, in which Tokyo Olympic and Paralympic Games will be held in Tokyo. For the next step, we will provide SaaS for public transportation for many satellite cities in Japan.

This public transportation application scenario is planned to be experimented in the third year of this project. So, the development and experiment plan is as follows:

- M01~M12: Requirement analysis and service/system design
- M13~M21: System and service development
- M21~M27: Experiment

## 5 Event Management – Tokyo Public Transportation (2)

The *Tokyo Public Transport application* is part of a set of applications that will use the CPaaS.io platform to manage fun and sports event. This sub-application focuses on ensuring reliable transportation of a wide range of vehicles in the city including for instance utility vehicles by public and private companies.

### 5.1 Application Description

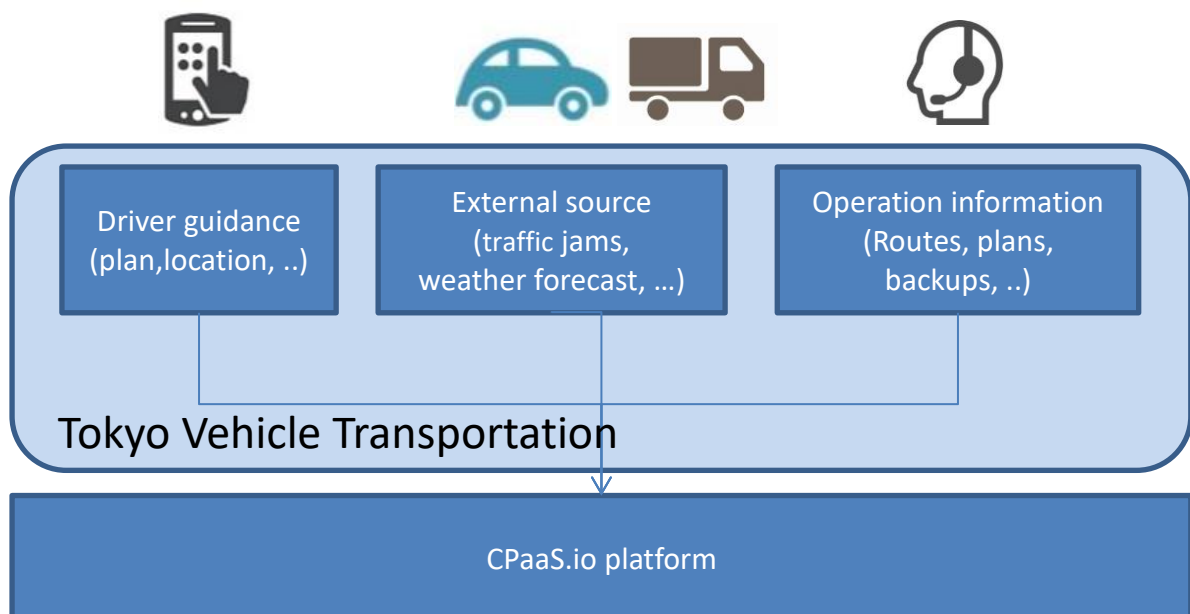
A city infrastructure for transportation includes buses, trucks, taxis, and utility vehicles by public and private companies. Reliable transportation systems require both, consumer-based and operator-based services. It is important for cities to manage the smooth coordination of both types of transportation services. It should orchestrate various types of city information in order to operate a wide range of vehicles in the city area.

In order to facilitate the efficient and reliable transport operation, it is necessary to incorporate vehicle locations, routes and plans, driver information and external information (such as weather, traffic jams) in a timely manner

Using the well-designed city information infrastructure, this trial case will be performed in Tokyo area.

### 5.2 Architecture

The following plan depicts the architecture plan overview of the vehicle transportation application of CPaaS.io. We will use the CPaaS.io platform for the underlying platform layer to support orchestration and integration of the collections of vehicle transportation and management services.



**Figure 11:** Vehicle Transportation: Initial version of the high level application architecture.

### 5.3 User Interface

The user interface will be provided as mobile web application that provides driver assistance of vehicle transportation. There will also be an integrated web application to incorporate the real-time operation, pre-defined routes and plans, real-time plan updates, and road information such as traffic conditions and weather forecast.

### 5.4 Data Sets

There are four types of data:

- Operation plan
- Operation logs
- External information in relation to operation: weather forecast, traffic conditions

| Operation plan              |                                                                        |
|-----------------------------|------------------------------------------------------------------------|
| <b>Detailed Description</b> | Routes, delivery time and place, contact addresses, driver information |
| <b>OGD or private data</b>  | private                                                                |
| <b>Personal Data</b>        | Contact addresses, driver information                                  |
| <b>Hosting</b>              | External                                                               |
| <b>Data Provider</b>        | Operation companies                                                    |
| <b>Format</b>               | JSON                                                                   |
| <b>Update Frequency</b>     | Daily (with some irregular updates)                                    |
| <b>Update Size</b>          | KB                                                                     |
| <b>Data Source</b>          | Operation companies                                                    |
| <b>Sensor</b>               | N/A                                                                    |
| <b>Number of Sensors</b>    | N/A                                                                    |

| Vehicle logs                |                                         |
|-----------------------------|-----------------------------------------|
| <b>Detailed Description</b> | Tracking data of vehicle transportation |
| <b>OGD or private data</b>  | Private                                 |
| <b>Personal Data</b>        | No                                      |
| <b>Hosting</b>              | External                                |
| <b>Data Provider</b>        | Driver of a vehicle                     |
| <b>Format</b>               | JSON                                    |
| <b>Update Frequency</b>     | Per minute                              |
| <b>Update Size</b>          | KB                                      |
| <b>Data Source</b>          | Smartphone of a driver                  |
| <b>Sensor</b>               | Embedded in a smartphone                |
| <b>Number of Sensors</b>    | 1 per smartphone                        |

| Traffic condition           |                                                            |
|-----------------------------|------------------------------------------------------------|
| <b>Detailed Description</b> | Traffic conditions and road status on transportation paths |
| <b>OGD or private data</b>  | private                                                    |
| <b>Personal Data</b>        | No                                                         |
| <b>Hosting</b>              | External                                                   |

|                          |                    |
|--------------------------|--------------------|
| <b>Data Provider</b>     | Service providers  |
| <b>Format</b>            | To be investigated |
| <b>Update Frequency</b>  | 1 per 10 minutes   |
| <b>Update Size</b>       | 10 KB              |
| <b>Data Source</b>       | Service providers  |
| <b>Sensor</b>            | Not known          |
| <b>Number of Sensors</b> | Not known          |

| Weather forecast            |                                          |
|-----------------------------|------------------------------------------|
| <b>Detailed Description</b> | Weather forecast of transportation paths |
| <b>OGD or private data</b>  | OGD                                      |
| <b>Personal Data</b>        | No                                       |
| <b>Hosting</b>              | External                                 |
| <b>Data Provider</b>        | Service providers                        |
| <b>Format</b>               | To be investigated                       |
| <b>Update Frequency</b>     | Hourly                                   |
| <b>Update Size</b>          | 10 KB                                    |
| <b>Data Source</b>          | Service providers                        |
| <b>Sensor</b>               | Not known                                |
| <b>Number of Sensors</b>    | Not known                                |

## 5.5 Deployment Plan

In the simplest case, vehicle location is tracked by the smartphone of the driver.

ACCESS has experience of implementing vehicle transportation planning and management systems.

In order to provide enhanced user experience, iterated cycles of field experiments and integration of city information data will be deployed using the CPaaS.io platform.

The deployment and improvement cycles of field tests will provide an important step-stone to provide robust and reliable operation-management services on top of a city information infrastructure platform. It will test the feasibility of a city information infrastructure for a wide range of operation services deployed in city areas.

This vehicle transportation use case is planned to be experienced in the third year of this project. The development and experiment plan is as follows:

- M01-M12: Requirement capturing and service/system design
- M13-M21: Service and system development, feasibility analysis of infrastructure configuration
- M22-M27: experiment and evaluation

## 6 Waterproof Amsterdam

The *rain proof Amsterdam application* is a single, but highly relevant application that will use the CPaaS.io platform to improve water management. Through smart, connected rain buffers, water supply peak shaving can be done in order to prevent streets, sidewalks and basements from flooding during heavy rainfall.

What makes this application interesting, is the ability to orchestrate the operation of multiple buffers simultaneously across a geographical network. This way, significant impact is expected which will actually address the existing bottlenecks of insufficient water processing capacity.

At the time of submitting this deliverable, The Things Network is negotiating with its local partners on a formal collaboration for development of this application. Waternet, the local water management company, is a required partner, as they are problem owner and provide data and infrastructure for both development and implementation of the application. If this collaboration does not succeed, The Things Network will not be in the position to further pursue this application. It will then choose to continue the back-up use application as input for the CPaaS.io project. This application is about smart parking of cargo trucks together with the Port of Amsterdam. The project plan for this application has been formalised already and there is commitment from the problem owner, so this alternative offers a reliable contingency plan. Clarity on the collaboration with Waternet is expected in October 2016, so minimal impact on other deliverables is expected.

## 6.1 Application Description

As research shows, extreme rainfall is becoming fiercer and will occur more frequently. Heavy rains are already causing occasional flooding and water damage in the city of Amsterdam. If the city does not adjust to the increasing intensity, the nuisance and damage will only increase in the future. Because the sewerage has not been designed for these enormous amounts of water, measures will have to be taken above ground in order to cope with peak loads. It has become clear that relatively simple large surfaces such as residential rooftops in the private domain may play a role in the water management.

### 6.1.1 Use case

IoT has the potential to help absorb rainwater at micro scale, slowly, and in a controlled manner through the use of water buffers – containers that collect, store and discharge beyond 1.000 litres of water per instance. These can be rooftops, rain barrels or underground tanks.

Roughly speaking, a solution has to deal with information from three different areas: 1. the private domain, 2. the public domain and 3. the weather / weather forecast. For an optimal contribution of the smart water management system to the existing water management infrastructure, these three domains need to be integrated, featuring an iterative self-learning system which takes into account to the limitations of the existing water management infrastructure. The buffers will be productised for the use in residential and commercial areas. Main premise is that an owner of a smart rain barrel can determine to what extent he / she wants to connect and share on the public network and how permission is given to be overruled by a third party such as Waternet (Amsterdam regional parastatal water management company) in case of predicted extreme rainfall or drought.

Below are a few design considerations for the smart buffer (hereafter called 'device'):

- Devices feature one or more sensors (e.g. to measure filling levels of buffer) and are equipped with a LoRa module (PHY layer) to communicate wirelessly using the open source LoRaWAN protocol.
  - Free to use and abundant LoRa connectivity is already present in the pilot areas
  - The LoRaWAN specs are suited for this use case, as it requires long battery life, little investments in infrastructure and only low data rates
- Devices feature an actuator that can be used to collect water into and release from the buffer. The actuation part of the application operates valves on the device which will control water inlet and water outlet.
- Device owners are able to individually fill/empty a buffer through an app

- Weather data can be fed into the system to determine the necessity of collecting or releasing water. This data may be supplemented with data from other sources such as sewerage filling degree, soil hydration, ground water levels and water infiltration rates, etc.
- The system will allow for a combination of autonomous and manual control – in case of emergencies, an authority like Waternet can take control of the actuators. Owners can however determine the degree of control they are willing to transfer (e.g. ranging from fully to semi-autonomous control)
- Machine learning may be applied for pattern recognition to optimise the network of buffers

This application will comprise both a software and hardware development, as it concerns a non-existent product and non-existing sources for data that serve as input and output for CPaaS.io.

### 6.1.2 Value Perspective

In the following tables we describe two values perspective. Table 4 lists the various user and stakeholder groups that benefit from the application and lists for each group the benefits they have by using the application. Table 5 in turns takes the perspective from the platform and lists for each key concept the benefit the CPaaS.io platform provides to the application.

**Table 4:** The benefits the application provides to the different user groups

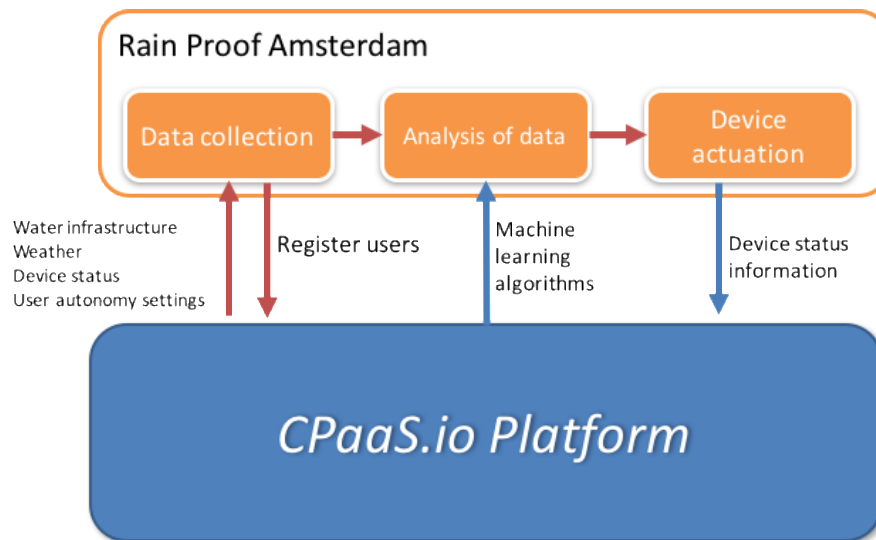
| User Value Perspective                                   |                                                                                                                                                                                                   |
|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Home owner or building owner who owns a buffer</b>    | Use the excess water collected as input for the 'grey water system' or for the garden/parks<br>Entertainment: gamification of water collection and release and visualised impact to bigger system |
| <b>Waternet – water management company</b>               | Centrally and (semi-)automatically managing of water release to sewerage infrastructure, taking pressure off the system.                                                                          |
| <b>Municipality – dept of infra and water management</b> | Fewer floodings and hinderance for properties and traffic<br>Best practice and benchmarking towards other cities                                                                                  |

**Table 5:** The benefits the CPaaS.io platform provides to the application

| Platform Value Perspective           |                                                                                                              |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Use of IoT Technologies</b>       | Centralised storage and control over data streams of various sources                                         |
| <b>Use of Open Governmental Data</b> | Ability to harmonise data from different governmental institutions<br>Ability to include location bound data |
| <b>Use of Linked Data</b>            | Not applicable                                                                                               |
| <b>Big Data Analysis</b>             | Ability to add analytics for pattern recognition and (semi-)automated operation of the buffer network        |
| <b>My Data Approach</b>              | Not applicable                                                                                               |

## 6.2 Architecture

Figure 12 shows an initial version of the high level components and their envisioned interaction with the CPaaS.io platform. It contains the following components:



**Figure 12:** Rain Proof Amsterdam: Initial version of the high level application architecture.

### 6.3 User Interface

The user interface for the end user (owner of the rain buffer) will be provided as mobile web application to operate the buffer and configure settings. There will also be a network operations centre web interface with aggregated, real time information about water situations (supply and demand) across the city and controls to operate the rain buffers.

Thirdly, the hardware components that need to be developed may also have a user interface. The rain buffers, whether a smart rooftop, rain barrel or underground container, all need to be equipped with hardware in order to gather data and potentially also actuate the device. The devices will be operated through digital commands, but a physical/mechanical fall back scenario also needs to be provided. This means a control panel will be part of the device.

### 6.4 Data Sets

We use the following types of data:

1. **Data from the private domain.**
  - a. The type of storage (rooftop, rain barrel, underground storage)
  - b. The per hour uptake and release capacity
  - c. The drainage destination (direct to sewerage, into garden soil)
    - i. with varying levels of soil humidity, for instance connected to soil sensors
  - d. Situation of buffers across a city, determining the density of the buffer network
2. **Data from the public domain:**
  - a. (Historical) water-related data: known areas of water flooding and other rainfall related issues, ground water levels
  - b. Water infrastructure: sewerage, capacity, bottlenecks, etc.
  - c. Data from weather forecast agencies

Depending on the availability of data to be stored in the CPaaS.io platform, the application may draw from this data to get data on local water infrastructure and weather.

| Weather data                |                                                              |
|-----------------------------|--------------------------------------------------------------|
| <b>Detailed Description</b> | Upcoming weather displaying periods of heavy rain or drought |
| <b>OGD or private data</b>  | OGD                                                          |
| <b>Personal Data</b>        | No                                                           |
| <b>Hosting</b>              | Platform                                                     |
| <b>Data Provider</b>        | KNMI – Dutch weather forecast agency                         |
| <b>Format</b>               | JSON, XML                                                    |
| <b>Update Frequency</b>     | Hourly                                                       |
| <b>Update Size</b>          | 10kb                                                         |
| <b>Data Source</b>          | Sensors                                                      |
| <b>Sensor</b>               | Water sensor                                                 |
| <b>Number of Sensors</b>    | unknown                                                      |

| Water buffer filling degree |                                                                         |
|-----------------------------|-------------------------------------------------------------------------|
| <b>Detailed Description</b> | Filling degree of a water buffer (rooftop, barrel, underground storage) |
| <b>OGD or private data</b>  | Private                                                                 |
| <b>Personal Data</b>        | Yes                                                                     |
| <b>Hosting</b>              | External                                                                |
| <b>Data Provider</b>        | TTN / Waternet                                                          |
| <b>Format</b>               | JSON                                                                    |
| <b>Update Frequency</b>     | Hourly or instantly with sudden increasing levels                       |
| <b>Update Size</b>          | 10b                                                                     |
| <b>Data Source</b>          | Sensors                                                                 |
| <b>Sensor</b>               | Water sensor                                                            |
| <b>Number of Sensors</b>    | 1 per buffer                                                            |

| Municipality water infrastructure |                                                                                      |
|-----------------------------------|--------------------------------------------------------------------------------------|
| <b>Detailed Description</b>       | Geographical data on water infrastructure depicting capacity and ground water levels |
| <b>OGD or private data</b>        | OGD                                                                                  |
| <b>Personal Data</b>              | No                                                                                   |
| <b>Hosting</b>                    | Platform                                                                             |
| <b>Data Provider</b>              | Waternet                                                                             |
| <b>Format</b>                     | XML                                                                                  |
| <b>Update Frequency</b>           | Hourly                                                                               |
| <b>Update Size</b>                | 1kb                                                                                  |
| <b>Data Source</b>                | Sensors, maps                                                                        |
| <b>Sensor</b>                     | Water sensor                                                                         |
| <b>Number of Sensors</b>          | unknown                                                                              |

## 6.5 Deployment Plan

The Things Network will develop this use case together with Water Management company Waternet. Waternet is launching a project called Rainproof Amsterdam, which pursues the objectives as stated above.

The project will bring together different partners in the value chain of this application, from interest groups who represent users and problem owners, to hardware engineers, product developers and software developers. The Things Network is currently in the process of aligning project partners. The development is an iterative process, so the below high level planning will show overlap between stages.

- M01~M12: Requirement analysis and service/system design
  - Align all stakeholders and receive contractual commitment from Waternet for the development of the use case (M04-M05)
  - Investigate areas in the city that dictate the highest urgency or best applicability for a pilot
- M03~M18: System and service development
  - Develop a smart, connected hardware prototype in lab setting.
  - Integrate prototype on The Things Network backend.
  - Set up a physical LoRa network in the pilot area by placing gateway on rooftop in pilot area
  - First and second release prototype
- M06~M28: Experiment and evaluation
  - Test of first and second release
  - Develop 10 more hardware prototypes to be deployed and tested in real life, outdoor setting.
  - Deploy prototypes in existing rain buffer projects in business district in Amsterdam (ZuidAs area).
  - Third release prototype

## 7 Yokosuka Emergency Medical Care

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The *Yokosuka Emergency Medical Care application* is a single, but highly relevant use application that will use the CPaaS.io platform to improve first aid management.

### 7.1 Application Description

To maintain the health and welfare of citizens, and to aid them in emergencies such as accidents or natural disasters in particular, emergency medical service is important.

Medical care professionals always race against time in emergency medical practice. YRP UNL has been developing an Emergency Medical Service Support System, which transmits the real-time location information of ambulances and images of patients to emergency hospitals via the tablet devices and IP cameras placed in ambulances. With this system, emergency wards of hospitals can learn about the conditions of the patients in advance and get ready to receive them smoothly. This Emergency Medical Service Support System has been used for two years in the city of Yokosuka, Japan.

Also, in the disaster-affected area, Triage Support System shares the triage levels of injured and sick people among medical professionals and rescue teams in real time and supports the smooth emergency rescue and medical care on site with active RFID tags that use Bluetooth Low Energy. Active RFID triage tags attached to injured and sick people allow medical care staff to learn the real-time state of the rescue site by constantly transmitting the triage levels of the patients by radio wave. The information is consolidated to give an overview of the rescue site.

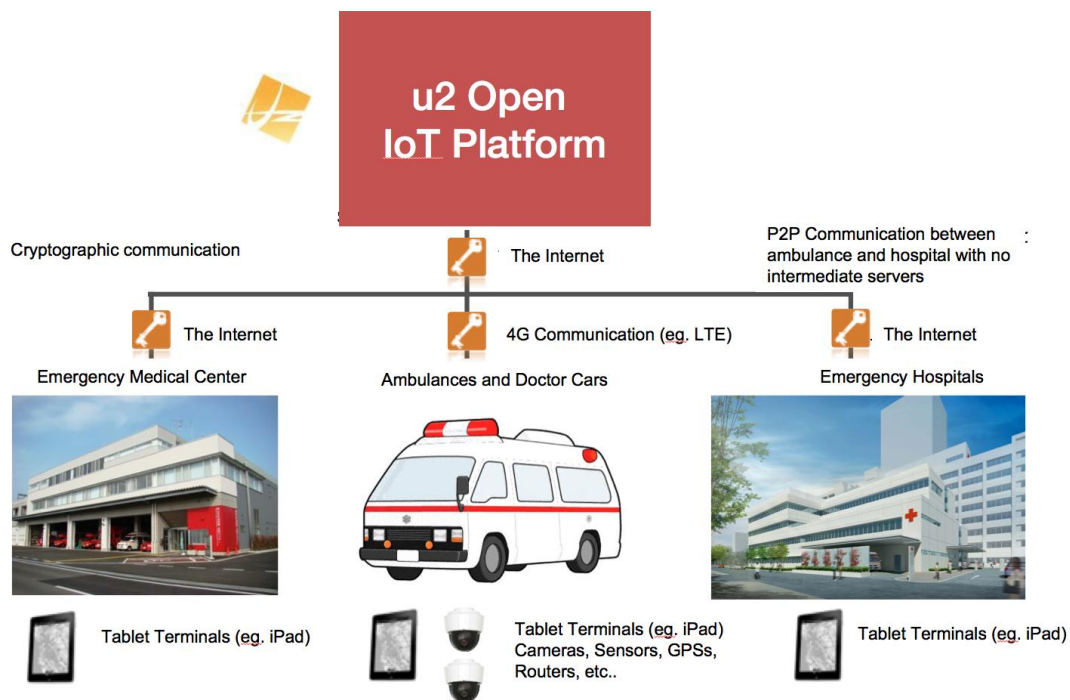
Emergency medical service is made possible by people of rescue teams on ambulances, at hospitals, and at dispatching headquarters, and by their teamwork. Information sharing among them is very important

to provide smooth emergency medical care. YRP UNL has been developing the system to collect the information from emergency hospitals and ambulances and share it among them and headquarters.

The Japanese-side consortium intends to enhance and extend the existing services to take full advantage of the new sensors, new networking technologies, and emerging CPaaS.io platform architecture to add new services and application to the existing system in this project for the benefit of citizens in Yokosuka and everywhere. The emergency medical staff of the City of Yokosuka where YRP's headquarters is located will help the consortium to carry out new prototype study there.

## 7.2 Architecture

The following figure is the architecture plan overview of Yokosuka Emergency Medical Service application of CPaaS.



**Figure 13:** Emergency Medical Care High Level Architecture

## 7.3 User Interface

The user interface for doctors and emergency medical service staff members will be provided as web application for mobile phones or tablets.



Figure 14: Visualisation of end-user interface of emergency medical information service

Doctors in emergency hospitals can see the location and movement of ambulances while they are heading to the hospitals.



Figure 15: large screen view of user interface of emergency medical information service

7.4 Data Sets

In our project, we will use data sets as summarized in the following table.

| Yokosuka Emergency Medical Care Data |                                     |
|--------------------------------------|-------------------------------------|
| Detailed Description                 | Emergency Medical Care Sensor data. |
| OGD or private data                  | Private Data                        |
| Personal Data                        | Some include Personal Data          |
| Hosting                              |                                     |

|                          |                                                                                                                                                                                                                         |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Data Provider</b>     |                                                                                                                                                                                                                         |
| <b>Format</b>            | ucR (ucode Relation) format (RDF + ucode in JSON format)                                                                                                                                                                |
| <b>Update Frequency</b>  | Real-time Update                                                                                                                                                                                                        |
| <b>Update Size</b>       | New real-time data is appended to (old) existing data.                                                                                                                                                                  |
| <b>Data Source</b>       | Emergency Service Team and Emergency hospitals of City of Yokosuka                                                                                                                                                      |
| <b>Sensor</b>            | All ambulances in Yokosuka City are equipped with sensors such as cameras (image sensors) and GPSs (location sensors).<br>In triage system, a triage tag works as a patient's location and medical condition indicator. |
| <b>Number of Sensors</b> | More than 100 sensors will be used.                                                                                                                                                                                     |

## 7.5 Deployment Plan

Presently, we are starting collaboration with Yokosuka City Office and all emergency hospitals in Yokosuka Cities. If our project can bring good results, we will make marketing effort for the deployment of our system in Yokosuka City. This is the first step. For the next step, we will extend the market to the Miura Peninsula, a region which includes Yokosuka City, Zushi City, Kamakura City, Miura City, and Hayama Town. These city and town halls are collaborating and they often adopt common municipal service systems as standard. After that, we will try to deploy our system across Japan.

This Yokosuka emergency medical service use case is planned to be experimented in the 2nd year of this project. So, the development and experiment plan is as follows:

- M01~M09: Requirement analysis and service/system design
- M10~M25: System and service development
- M16~M21: Experiment

## 8 Summary

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In this deliverable we have described 5 application scenarios and their requirements that will drive the design and implementation of the CPaaS.io platform. For each of the application scenarios we have provided a high level architecture, where available a user interface, a list of data sets and a deployment plan.

The architecture sketches an initial idea of platform interactions and is meant to be quickly iterated in the platform requirements analysis phase. The architectures are expected to change considerably throughout the course of the project. For instance we expect that some of the components that are currently considered to be part of the application may indeed be offered by the platform. Here is a summary of initial observations across all the application architectures:

- All applications are concerned with **data collection**. While the data collection subsystems certainly contain application specific components, e.g. reading data from wearable sensors, we would expect that the platform provides a capability to ingest data from various sources. At the very minimum level however the platform must offer a capability to register data sources so that data can be used across applications.
- Most of the applications use a data **analysis component**. This component varies strongly from application to application with respect to both how data is processed and how fast it needs to be

processed. Processing may include simple analysis using aggregations of the original data or involve complex machine learning algorithms. With respect to the speed of processing, some applications such as the Enhanced User Experience, use machine learning algorithms and ultimately require low latencies which may require the need of complex event processing infrastructures. As a next step we will identify what infrastructure components and possibly analytic toolboxes should be provided by the platform.

- Some of the applications such as Waterproof Amsterdam require **actuation services**. There may be a case that the platform provides a capability to manage access to various actuators distributed in the city.
- Most of the use cases require that the platform **manages and registers users**. For instance the application scenarios within the event management use case provide personalized services.

As of now we have identified 24 data sets. Across the applications the data is diverse including structured sensor data, maps and video data originating from both open governmental and private data sources. We also have varying data formats including for instance JSON and CSV and data that is already provided as RDF triples as in the case of the Tokyo Public Transportation application scenario or the Emergency Medical Care application. This means that we need to transform or annotate non-RDF data sources in order to facilitate semantic data integration. Update frequencies also vary significantly between 16ms and yearly updates which shows that the data needs to be ingested in high rates.

The deployment plans of the individual applications show the progress of the planning and implementation of the application scenarios. For the Enhanced User Experience use case, an initial data collection in order to evaluate the potential concrete services has already been executed. The services for the Sapporo Visitor Experience will be provided in February 2017. The deployment of the Tokyo Public Transportation application scenario is done in collaboration with 50 members including key stakeholders and global IT companies such as Fujitsu, NEC, Google and Microsoft and planned for the third year of the project. The Waterproof Amsterdam use case addresses a pressing problem for the city of Amsterdam and progress is ensured as TTN is seeking collaboration with Waternet as a key partner targeting a first prototype in Q4 2016. Finally, for the Emergency Medical Care use case there is already a prototype that will be deployed in the second year of the project.

## Appendix A - Backlog of User Stories

In this appendix we provide for each application an initial set of requirements as a backlog of user stories. The requirements are consumed by the requirements analysis task of WP 3 and also constitute a list of epics that can further be broken in smaller user stories that will be implemented during the implementation phase. The requirements have the format as detailed in Table 6. The list of requirements for the Sapporo Visitor Experience will be provided in conjunction with the deliverable D2.2.

**Table 6:** Field description of backlog items

| FIELD              | DESCRIPTION                                                                                                                                                  |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ID</b>          | Each requirement is uniquely identified of the form <application_id>-<number>, e.g. for requirement 1 of the enhanced user experience application: EUE-1     |
| <b>Description</b> | The description is the intent of the requirement. It is a statement about what the system has to fulfil according to the rationale.                          |
| <b>Rationale</b>   | The rationale is the reason behind the requirement's existence. It explains why the requirement is important and how it contributes to the system's purpose. |

|                      |                                                                                                                                                                                                                                   |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Fit Criterion</b> | A quantification or measurement to assess to which extent the original requirement has been eventually implemented/taken into account within the system. In some cases only the criteria used for quantification may be provided. |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Event Management - Enhanced User Experience

| ID            | Description                             | Rationale                                                                                                                                                                               | Fit Criterion                                            |
|---------------|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| <b>EUE-1</b>  | Data collection infrastructure          | All services require the collection of real-time data                                                                                                                                   | Existence of a data collection infrastructure            |
| <b>EUE-2</b>  | Metadata management                     | Sensor data can only be processed if it is known how sensors were deployed , operated, to which user, event etc. the data relates etc.                                                  | Method for collecting metadata and enriching sensor data |
| <b>EUE-3</b>  | Visualization of personal data          | The visualization component is the front end towards end user that provides him/her access to a record of his/her event experience.                                                     | Visual data access for end user                          |
| <b>EUE-4</b>  | Data processing                         | Data such as excitement levels require processing of raw sensor data                                                                                                                    | Availability of analytical services                      |
|               | User Registration and Management        | This application offers personalized services.                                                                                                                                          | Existence of user registration management component      |
| <b>EUE-5</b>  | Control over data collection            | This allows users to be empowered to make a decision between what data is collected and what services they can use                                                                      | UI for controlling use of data                           |
| <b>EUE-6</b>  | Synchronized clocks between IoT devices | Time synchronization is required so that sensor data from different sensors can be correlated by time                                                                                   | Maximum clock drift                                      |
| <b>EUE-7</b>  | Efficient query of user data            | Personalized visualizations require efficient retrieval of media data and processed data from sensor streams                                                                            | Query latency in ms                                      |
| <b>EUE-8</b>  | Guarantees for base functionality       | When local (BT links, wireless links to local base stations) or backend connectivity (cloud) is lost, the application must still provide value to the end user                          | User value                                               |
| <b>EUE-9</b>  | Data completeness                       | When local (BT links, wireless links to local base stations) or backend connectivity (cloud) is lost, it must be ensured that data loss is avoided in order to ensure service operation | Percentage of lost packets                               |
| <b>EUE-10</b> | Data sharing                            | Users may want to share their data via various channels using various platform instances                                                                                                | Availability of data sharing mechanisms                  |

|               |                                              |                                                                                                                                                           |                                                                                  |
|---------------|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <b>EUE-11</b> | Semantic data integration                    | Data sharing between diverse systems can only efficiently be achieved if the semantics of the data are machine readable                                   | Ontologies for data to be shared                                                 |
| <b>EUE-12</b> | Scale to personal data of millions of people | The application will collect sensor and media data for millions of people. The platform needs to be able to efficiently ingest, store, process user data. | Number of data points ingested per second, processing latency, storage footprint |

## Event Management – Tokyo Public Transportation (YRP)

| ID            | Description                           | Rationale                                                                                                                                                                                               | Fit Criterion                                                                                       |
|---------------|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| <b>TPT -1</b> | Automated data collection mechanism   | Public transportation data tends to update frequently. Efficient update of the data requires an automatic data collection system (even for static data such as time tables and maps of stations).       | Existence of automatic data collection system for public transportation.                            |
| <b>TPT -2</b> | Multi-lingual data                    | Foreign tourists may not have Japanese language skills and need multilingual information on public transportation.                                                                                      | Ratio of multilingual information in all information.<br>Existence of automatic translation system. |
| <b>TPT -3</b> | Live data collection and distribution | This use case deals with real-time operation information of public transportation such as trains, busses, an air planes. It is necessary to provide automatic data collection mechanism for these data. | Existence of automatic collection system and distribution system of live operation data.            |
| <b>TPT -4</b> | Data format and API standardization   | Since public transportation is operated by many private and public sectors, data distribution can only be achieved efficiently based on standard data formats and APIs.                                 | Existence of standard data format and standard API specifications.                                  |
| <b>TPT -5</b> | Secure and robust communication       | Public transportation is a critical social infrastructure. Communication among players of the public transportation open data system should be secure and robust.                                       | Existence of cryptographic communication mechanism and fault tolerance mechanism.                   |
| <b>TPT -6</b> | Developers' Site                      | For the open innovation with public transportation open data, it is necessary to provide an open development environment on the platform called "developers site".                                      | Existence of developers' site of Tokyo Public Transportation Open Data.                             |

|                |                                        |                                                                                                                                                                           |                                                   |
|----------------|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| <b>TPT -7</b>  | Data License                           | Good data license is necessary for the deployment of the data. For the data license, the balance between data holders' merits and users' merits is essentially important. | Existence of open data license.                   |
| <b>TPT -8</b>  | Third party apps                       | Open data is a method for the collaboration between data holders and application developers.                                                                              | Number of third party apps or software.           |
| <b>TPT -9</b>  | Services for the physically challenged | Contribution to improve quality of life of the physically challenged and the elder in moving in Tokyo is one of the most important objectives.                            | Number of services for the physically challenged. |
| <b>TPT -10</b> | User feedback collection               | User feedback is necessary for the evaluation of the use case experiment.                                                                                                 | Number of user feedbacks.                         |

### Waterproof Amsterdam (TTN)

| ID          | Description                   | Rationale                                                                                                                                                                                                                                                         | Fit Criterion                                          |
|-------------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| <b>WA-1</b> | Data collection               | The application collects and stores data from different sources, including sensors and static cadastral and infrastructural information                                                                                                                           | Existence of data collection infrastructure            |
| <b>WA-2</b> | Data processing               | A rain buffer contains a set of standard parameters, such as type, location, capacity. These parameters should be complemented with lifetime usage data, such as times/open close, amount buffered/released, active/inactive, filling level, battery status, etc. | Existence of data collection infrastructure            |
| <b>WA-3</b> | User hierarchies              | Different user groups can exert different levels of control over a device, so a hierarchy between groups is established                                                                                                                                           | Definition of user groups, definition of access rights |
| <b>WA-4</b> | Manual control over device    | Rules are set by individual users which determine the level of control over their own device. Users may have different preferences.                                                                                                                               | UI where users can set their preferences               |
| <b>WA-5</b> | Automated control over device | Circumstances need to be defined, based on the various data sources, in which device control for user groups are suspended and delegated to a central user/authority                                                                                              | Presence of algorithms                                 |
| <b>WA-6</b> | Adaptive control levels       | The rules and circumstances may automatically trigger control levels as they occur                                                                                                                                                                                | Control levels defined                                 |

|              |                                |                                                                                                                                              |                                                 |
|--------------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| <b>WA-7</b>  | Visualisation of device status | Users can access information via a graphical front-end that displays a record of the rain buffer status, such as filling level               | UI that can be accessed by user                 |
| <b>WA-8</b>  | Operator control centre        | Data can be aggregated per geographical area in order to monitor the status of and trigger actions on a number of devices altogether         | UI for network operator                         |
| <b>WA-9</b>  | Push notifications             | Circumstances may trigger communication of predefined messages to users such as push notifications.                                          | Reception of notifications                      |
| <b>WA-10</b> | User feedback collection       | Prompts in the application can be used to manually collect user feedback.                                                                    | Collection of feedback                          |
| <b>WA-11</b> | Data harmonisation             | Data collected from the various sources needs to be useable and interpretable by the application                                             | Ontologies for data to be shared                |
| <b>WA-12</b> | Physical network               | Data transmission is modulated with LoRa, operated by The Things Network network server. The application needs to be integrated with TTN API | Data received in application server via TTN API |
| <b>WA-13</b> | Multicast messaging            | Bulk downlink messages may be sent to control large quantities of actuators at the same time.                                                | Messages sent from control centre               |

## Yokosuka Emergency Medical Care (YRP)

| ID           | Description                                                      | Rationale                                                                                                                                                                                                            | Fit Criterion                                    |
|--------------|------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| <b>YEM-1</b> | Data collection                                                  | This use case deals with real-time operation information of such as patients' conditions, and position of ambulances. It is necessary to provide automatic data collection mechanism for these data.                 | Existence of automatic collection system         |
| <b>YEM-2</b> | Consensus for using medical and personal data by city government | The data may have direct or indirect relations with personal information of patients. Before the trial, we need to have formal consensus with city government on the basis of official guidelines for personal data. | Agreement with Yokosuka city government          |
| <b>YEM-3</b> | Privacy protection of patients                                   | We must provide strong data protection mechanism such as secure communication and secure storage.                                                                                                                    | Existence of privacy data protection mechanism   |
| <b>YEM-4</b> | Low-effort for emergency medical team staffs                     | Emergency medical team staffs are doing very hard work for taking care of patients. It is impossible for them to                                                                                                     | Existence of low-effort mechanism and evaluation |

|             |                          |                                                                                                                                 |                          |
|-------------|--------------------------|---------------------------------------------------------------------------------------------------------------------------------|--------------------------|
|             |                          | spend time and effort for the operation of this system. We must realize a good work-flow and user interface for the team staff. | score by the team staffs |
| <b>WA-6</b> | User feedback collection | User feedback is necessary for the evaluation of the use case experiment.                                                       | Number of user feedbacks |